Math728J – Homework 2

1. Let $\mathbf{A} \in \mathbb{R}^{N\times N}$ be spd. Assume that there are exactly $k \leq N$ distinct eigenvalues of $\mathbf{A}$. Show that the CG method will terminate in at most k iterations.

2. Let $\mathbf{A} \in \mathbb{R}^{N\times N}$ be spd. Show the following relationships hold for the CG method with $\mathbf{x}_0 = 0$:

   a. $\mathbf{r}_k^T \mathbf{p}_{k+1} = \|\mathbf{r}_k\|_2^2$

   b. $\mathbf{p}_{k+1} = \|\mathbf{r}_k\|_2^2 \sum_{j=0}^{k} \frac{\mathbf{r}_j}{\|\mathbf{r}_j\|_2^2}$

   c. $\|\mathbf{p}_{k+1}\|_2^2 = \|\mathbf{r}_k\|_2^4 \sum_{j=0}^{k} \|\mathbf{r}_j\|_2^{-2}$

   d. $\mathbf{p}_k^T \mathbf{p}_j = \frac{\|\mathbf{r}_{k-1}\|_2^2}{\|\mathbf{r}_{j-1}\|_2^2} \|\mathbf{p}_j\|_2^2 \quad \text{for} \quad j \leq k$

   e. $\|\mathbf{x}_{k+1}\|_2 \geq \|\mathbf{x}_k\|_2$

   f. $\|\mathbf{r}_k\|_2 \leq \|\mathbf{p}_{k+1}\|_2$

3. Let $\mathbf{A} \in \mathbb{R}^{N\times N}$ be spd.

   a. Show that
   $$\frac{\|\mathbf{r}_k\|_2}{\|\mathbf{r}_0\|_2} \leq \sqrt{\kappa_2(\mathbf{A})} \frac{\|\mathbf{x}^* - \mathbf{x}_k\|_A}{\|\mathbf{x}^* - \mathbf{x}_0\|_A}$$

   b. If $\kappa_2(\mathbf{A}) = O(N)$, give a rough estimate of the number of CG iterates with $\mathbf{x}_0 = 0$ required to reduce the relative residual to $O(1/N)$.

4. (**Programming Exercise**) Let $\Omega = (0,1)^2$. Considser numerical solution of the boundary value problem

$$\begin{cases} -u_{xx}(x,y) - u_{yy}(x,y) + b(x,y)u(x,y) = f(x,y), & (x,y) \in \Omega \\ u(x,y) = 0, & (x,y) \in \partial\Omega \end{cases} \tag{1}$$

where $b(x,y) = 1$. We let $u(x,y) = 10.0xy(1-x)(1-y)$ and then the right hand side f is determined correspondingly. We consider an uniform square grid with grid points (x_i, y_j) which are chosen to be equidistant:

$$x_i = ih, \quad y_j = jh, \quad i,j = 0,1,\cdots,N \quad (h = 1/N)$$

and the center difference scheme:

$$\frac{-U_{i-1,j} - U_{i,j-1} + 4U_{i,j} - U_{i+1,j} - U_{i,j+1}}{h^2} + b_{i,j}U_{i,j} = f_{i,j},$$

$$i,j = 1,2,\cdots,N-1,$$

where $b_{i,j} = b(x_i, y_j)$ and $f_{i,j} = f(x_i, y_j)$, and

$$U_{i,j} \approx u(x_i, y_j), \qquad i, j = 1, 2, \cdots, N-1$$

$$U_{i,0} = U_{i,N} = U_{0,i} = U_{N,i} = 0, \qquad i, j = 0, 1, \cdots, N.$$

Let $\mathbf{x}$ be the vector consisting of unknowns $U_{i,j}$ $(i, j = 1, 2, \cdots, N-1)$ ordered row-wise (from left to right) and upward. Then, we get a linear system such as

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

where $\mathbf{A} \in \mathbb{R}^{(N-1)^2 \times (N-1)^2}$ is a sparse matrix and $\mathbf{b} \in \mathbb{R}^{(N-1)^2}$. For $N = 10, 20, 40, 80$, determine an approximate solution of (1) using the Conjugate Gradient method, where the iteration is stopped when either more than $Kmax = 500$ iterations are reached or the residual $\|\mathbf{r}_k\|_2 < \epsilon \|\mathbf{b}\|_2$ with $\epsilon = \dfrac{h^2}{100}$. Choose $\mathbf{x}_0 = 0$ as the initial value for the iterations. Hand in your codes with some explanations. Provide the number of required iteration steps k, and the errors $\max\limits_{i,j=1,2,\cdots,N-1} |U_{i,j} - u(x_i, y_j)|$ in a table. Also plot the history diagram of number of iterations k *vs.* the relative residual $\|\mathbf{r}_k\|_2 / \|\mathbf{b}\|_2$ in a semi-log scale for the $N = 40$ case using Matlab (the command *semilogy*).

5. (**Programming Exercise**) Re-do Problem 4 using preconditioned Conjugate Gradient method with the symmetric Gauss-Seidel preconditioner $\mathbf{B}_{SGS} = (\mathbf{D} + \mathbf{L}^T)^{-1} \mathbf{D} (\mathbf{D} + \mathbf{L})^{-1}$. (Compressed Sparse Row format can be used for data storage scheme).